Source-function order on the actual weighted-Proj curve

Formal proof packet

28 October 2026

This packet concerns the actual scheme $X = \mathrm{Proj}(\mathrm{degreeSubmodule}(a_3, a_2, a_1, a_0))$ over a field K. It reports conditional formal declarations, not a new DLP run or cost result.

Source and hypotheses

The existing source-shaped common 2+2 function $N/B - y$ is embedded in $K(X)$ under separability of the quintic. The already checked $\mathrm{IsIntegral}(X)$ theorem uses separability; the checked $\mathrm{IsLocallyNoetherian}(X)$ theorem requires only that K be a field. These are Mathlib's typeclass prerequisites for $\mathrm{ord}_z(f)$.

New declarations

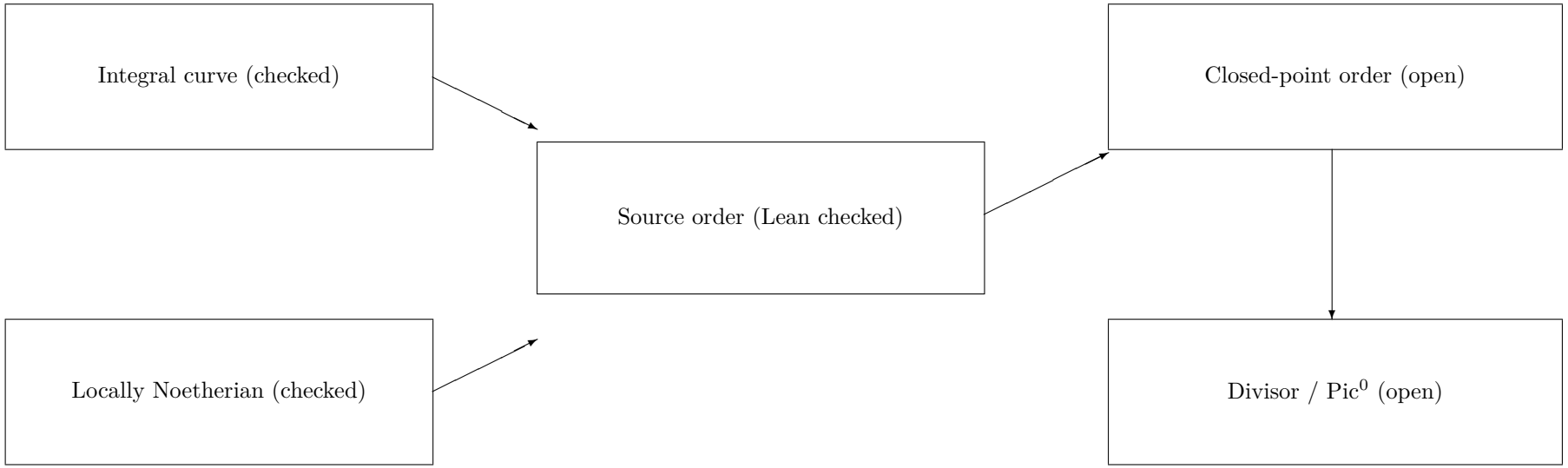
- `sourceOrder` specializes ord_z to the source function at any point $z \in X$.
- `sourceOrder_zero_of_coheight_ne_one` records the API's zero value when $\mathrm{coheight}(z) \neq 1$.
- `sourceOrder_eq_ordHom_of_coheight_one` identifies it with Mathlib's stalk valuation when $\mathrm{coheight}(z) = 1$.

Pinned Lean 4.34.1 exited zero. Direct axiom prints for all three new declarations list only `propext`, `Classical.choice`, and `Quot.sound`; the compiler receipt is retained with the source.

Open gates

Neither closed-point coheight nor nonvanishing of the source function is inferred. Finite support, the infinity pole, the principal divisor, the Pic^0 equation, and compiled Rust correspondence remain open. The CSV lists every gate separately; a zero returned by ord is not by itself an order calculation.

Dependency diagram



The source function's nonvanishing, finite support, infinity pole, and compiled semantics are separate obligations.